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Section 1: SIMPLE LINEAR REGRESSION

Question #01

Given

$$x = [1, 2, 3, 4, 5]$$

$$y = [2, 3, 5, 7, 11]$$

(a) Compute Slope (β_0) and Intercept (β_0)

$$n = 5$$

$$\sum x = 1 + 2 + 3 + 4 + 5 = 15$$

$$\sum y = 2 + 3 + 5 + 7 + 11 = 28$$

$$\sum xy = (1 \cdot 2) + (2 \cdot 3) + (3 \cdot 5) + (4 \cdot 7) + (5 \cdot 11) = 2 + 6 + 15 + 28 + 55 = 106$$

$$\sum x^2 = 1 + 4 + 9 + 16 + 25 = 55$$

Slope β_1

$$\beta_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = \frac{(5 \cdot 106) - (15 \cdot 28)}{(5 \cdot 55) - (15)^2} = \frac{110}{50} = 2.2$$

Intercept β_0

$$\beta_0 = \frac{\sum y - \beta_1 \sum x}{n} = \frac{28 - 2.2 \times 15}{5} = \frac{28 - 33}{5} = \frac{-5}{5} = -1$$

REGRESSION EQUATION: $\hat{y} = -1 + 2.2x$

(b) Predict y when $x = 6$

$$\hat{y} = -1 + 2.2 \times 6 = -1 + 13.2 = 12.2$$

(c) Compute R^2

$$\sum y^2 = 4 + 9 + 25 + 49 + 121 = 208$$

$$\bar{y} = \frac{\sum y}{n} = \frac{28}{5} = 5.6$$

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$$\begin{aligned}
 SS_{\text{Total}} &= \sum y^2 - n\bar{y}^2 \\
 &= 208 - 5(5.6)^2 \\
 &= 208 - 156.8 \\
 &= 51.2
 \end{aligned}$$

Compute Predicted Values

$$\hat{y} = [-1 + 2.2 \times 1, -1 + 2.2 \times 2, -1 + 2.2 \times 3, -1 + 2.2 \times 4, -1 + 2.2 \times 5]$$

$$\hat{y} = [1.2, 3.4, 5.6, 7.8, 10]$$

$$\begin{aligned}
 SS_{\text{res}} &= \sum (y_i - \hat{y}_i)^2 = (2 - 1.2)^2 + (3 - 3.4)^2 + (5 - 5.6)^2 + (7 - 7.8)^2 + (11 - 10)^2 \\
 &= 0.64 + 0.16 + 0.64 + 0.36 + 1 \\
 &= 2.8
 \end{aligned}$$

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{Total}}} = 1 - \frac{2.8}{51.2} \approx 1 - 0.0547 = 0.9453$$

Question #02

Given

$$Y_{\text{actual}} = [3.6, 8, 11]$$

$$Y_{\text{Predicted}} = [2.8, 5.5, 8.2, 11.1]$$

Compute MSE

$$\begin{aligned}
 MSE &= (1/n) \sum (Y_{\text{actual}} - Y_{\text{predicted}})^2 \\
 &= \frac{(3 - 2.8)^2 + (8 - 5.5)^2 + (11 - 8.2)^2 + (11 - 11.1)^2}{4} = \frac{0.04 + 0.25 + 0.04 + 0.01}{4} \\
 &= \frac{0.34}{4} = 0.085
 \end{aligned}$$

Compute RMSE

$$RMSE = \sqrt{0.085} \approx 0.2915$$

$$\text{For } x=1 : 0.2$$

$$\text{For } x=2 : 0.5$$

$$\text{For } x=3 : -0.2$$

$$\text{For } x=4 : -0.1$$

Errors are less than 1 in magnitude,
results small MSE value 0.085

Question #03

Answer: Model B performs better because it has lower MSE 2.3 as compared to Model A 4.5.

MSE measures the avg. squared difference between predicted values and actual values, so the lower MSE indicates better model performance with smaller prediction errors.

Section 2: MULTIPLE LINEAR REGRESSION

Question #04

Given: $x_1 = [1, 2, 3, 4]$

$x_2 = [2, 3, 4, 5]$

$y = [3, 6, 9, 12]$

(a)

Regression Equation: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$

(b)

Compute β using Normal Equation:

$$\beta = (X^T X)^{-1} X^T Y$$

Matrices:

$$X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{bmatrix}, \quad Y = \begin{bmatrix} 3 \\ 6 \\ 9 \\ 12 \end{bmatrix}$$

• first column = 1

for intercept β_0

• second column = x_1

• Third column = x_2

2. Compute $X^T X$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 14 \\ 10 & 30 & 42 \\ 14 & 42 & 54 \end{bmatrix}$$

3. compute $X^T Y$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ 9 \\ 12 \end{bmatrix} = \begin{bmatrix} 30 \\ 90 \\ 126 \end{bmatrix}$$

4. solve $(X^T X) \beta = X^T Y$

$$\begin{bmatrix} 4 & 10 & 14 \\ 10 & 30 & 42 \\ 14 & 42 & 54 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 30 \\ 90 \\ 126 \end{bmatrix}$$

$\Rightarrow Y = 3 \cdot X_1$ [3, 6, 9, 12] are exact $3 \cdot (1, 2, 3, 4)$

$$X_2 = X_1 + 1$$

Relation

$$Y = 0 \cdot \beta_0 + 3 \cdot X_1 + 0 \cdot X_2$$

5. Solving the System:

using Row Reduction

$$\beta = (X^T X)^{-1} X^T Y$$

$$\text{solution: } \beta_0 = 0, \beta_1 = 3, \beta_2 = 0$$

$$Y = 0 + 3X_1 + 0X_2$$

$$= 3X_1 \Rightarrow \text{which matches given } Y = [3, 6, 9, 12]$$

• Predict y for $x_1 = 5$, $x_2 = 6$

By Regression Equation

$$\hat{y} = 0 + 3(5) + 0(6) = 15$$

$$\text{where } y = y_1 + y_2 \text{ but } y = 3X_1$$

• X is $3X_1$ • β is $[0, 3, 0]^T$

SOLUTIONS:

$$(a) X = 0 + 3X_1 + 0X_2$$

$$(b) \beta = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$$

$$(c) \hat{y} = 3 \cdot 5 = 15$$

Question #05: Reasoning:

No, a higher R^2 does not always mean a 'better model'.

- Adding variable can overfit the data using adjusted R^2 or cross validation.

Question #06:

Effect:

multicollinearity inflates variance of β -coefficients.

Detection:

High variance Inflation Factor ($VIF > 5$) estimates.

Solution:

- Remove correlated predictors or use Ridge Regression.
- Collection of more data.

Section 3: POLYNOMIAL REGRESSION

Question #01:

Given:

$$x = [1, 2, 3, 4]$$

$$y = [2, 5, 10, 17]$$

(a) Transformed Equation:

Degree = 02

$$x_{poly} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix}, \quad y = \begin{bmatrix} 2 \\ 5 \\ 10 \\ 17 \end{bmatrix}$$

(b) Solve for β

$$\beta = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

where $y = 1 + x^2$

(c) Predict y for $x=5$:

$$\hat{y} = 1 + 0.5 + 1.25 = 2.75$$

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Question #08:

Explanation

Model B (degree=4) overfits the training model, leading to higher test error than Model A (degree=2).

Question #09:

reasoning

Overfitting occurs when the model fits noise. Regularization penalizes coefficients to improve generalization.

Section 4: LOGISTIC REGRESSION

Question #10:

Given:

$$z = -2 + 0.8x$$

$$x = 3$$

Compute Probability:

$$p = \frac{1}{1 + e^{-(z)}} = \frac{1}{1 + e^{-0.4}} \approx 0.5987$$

Question #11:

$$Y_{\text{actual}} = [1, 0, 1, 1]$$

$$Y_{\text{predicted}} = [0.9, 0.2, 0.8, 0.7]$$

log loss:

$$-\frac{1}{4} [\ln(0.9) + \ln(0.8) + \ln(0.7) + \ln(0.2)] \approx 0.2271$$

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Question #12:

Gradient of log loss:

$$\nabla \beta l = x^T (\sigma(z) - y)$$

Question #13:

Gradient Descent Update:

- Compute $\sigma(z)$ for each row

- Update β

$$\beta_{\text{new}} = \begin{bmatrix} -1 \\ 0.5 \end{bmatrix} - 0.01 \begin{bmatrix} -0.3288 \\ -0.3023 \end{bmatrix} \approx \begin{bmatrix} -0.9967 \\ 0.5030 \end{bmatrix}$$
